

Sample End Semester Question Paper - 2025

Subject: Python Programming | Code: CS201 | Semester: 2

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. State the difference between a list and a tuple.
2. What is a dictionary?
3. What does try-except do?
4. What is the `__init__` method?
5. Name one built-in string method used for searching.
6. Name two features that make Python popular.
7. State two mutable data types in Python.
8. What is a lambda function?
9. How are mutable objects passed to functions in Python?
10. What does the `len()` function do?

PART B - Long questions (10 x 8 = 80 Marks)

1. Write recursive programs: factorial, Fibonacci and Tower of Hanoi.
2. Explain sets and tuples and their operations with examples.
3. Explain Python data types, variables and operators. Write a program using them.
4. Explain functions in Python: arguments, return values, scope and lambda functions with examples.
5. Explain lists: creation, operations and list comprehensions with programs.
6. Explain dictionaries and file handling in Python with example programs.
7. Explain exception handling in Python with examples.
8. Explain OOP in Python: classes, objects, inheritance, encapsulation and polymorphism with a program.
9. Explain strings and string methods with examples.
10. Explain modules and packages. Demonstrate with a small project structure.